

NATHAN MANLEY

Cork, Ireland | First-Year Engineering Student at UCC | Embedded Systems and Space Enthusiast
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More information on my projects and experience available on my website.

EDUCATION

- University College Cork** — *1st Year Engineering — First Class Honours (1.1) | GPA: 3.75/4* 2025 - 2029
- Quercus Talented Student Programme Innovation & Entrepreneurship Scholar** — *Fully funded scholarship, awarded as recognition for entrepreneurial and innovative endeavours both on and off campus.* 2026 - 2029
- Glanmire Community College** — *Leaving Certificate — 589/625 Points* 2025
- Subjects:** Applied Maths, Physics, Technology, Maths, English, History, Irish, French

WORK EXPERIENCE

- Logitech** — *Software UX Designer* 2026 - Present
- Creating and designing plugins for Logitech MX Keypad.
- Logitech** — *Software UX Design Intern* Summer 2026
- UX Design Explorations for upcoming PWS POVs.
- ATMac** — *Repair Technician* Summer 2023/2024
- Diagnosed and repaired various Apple laptops and desktops.
 - Enhanced technical troubleshooting and component level repair skills.

TECHNICAL / ENGINEERING EXPERIENCE

- National SDG Hackathon**
- Winner of National SDG Hackathon, UCC and Best Pitch Award, winning €4K in prize money.
 - Developed Remento, an app for early-stage dementia tracking/detection for families of the elderly.
- HackEurope** 2026
- Developed WindSite, utilising Claude, Python, and React Web Development.
 - WindSite is an intelligent platform that integrates a modular node system flow chart to make offshore wind turbine farm planning easier and smarter.
- Patch** 2024
- Selected as one of 30 innovators across Europe to complete the program in Dublin.
 - Co-founded Gymficient, a platform providing actionable analytics to gym owners for machine acquisition and maintenance.
 - Developed backend logic and analytics modules using OpenCV, in C++ and Python.
- UCC Rocketry and Space Exploration Society** — *Payload and Hardware Avionics Team Member* 2025–2026
- Programming camera telemetry and payload systems for MACH '26. AV Bay 3D Designer for EuRoC '26.
- CanSat** — *Lead 3D Can Design* 2024
- Designed core satellite structure using Autodesk Fusion 360.
 - Optimized housing for electronics and produced parts on my personal 3D printer.
- CanSat** — *Lead Programmer* 2023
- Implemented real-time telemetry and data processing in C++.
 - Built post-processing algorithms and visualized results for presentation in the competition.

INDEPENDENT PROJECTS / ACHIEVEMENTS

- Telemetry Parsers: Captures and interprets UDP data from simulators like Gran Turismo 7 for integration with microcontrollers and dashboards. Various modules were programmed in C++, Python and Javascript.
- School Awards: Environmental and Innovation “Diograis” awards while attending GCC.

VOLUNTEERING

CoderDojo Glanmire

Joined as a programmer in 2014, receiving free mentorship. I became a mentor in 2023 to volunteer my time to teach today's young learners C++, Python, and HTML.

Gaisce

Bronze Gaisce award.

2023

INTERESTS

- Technical: Microcontrollers (ESP32/ESP8266), Embedded Systems, 3D Printing (Ender 3 V2).
- Personal: Weightlifting, Formula 1, Drifting.
- Academic: Electrical, Electronic & Mechanical Engineering. Rocketry and CanSat.

References available upon request